

7.9 PreQuiz: Exponential and Logarithmic Word Problems (calculator)

1. A bacteria culture begins with 200 cells. The number of cells N , t hours after the start of the experiment, is modelled by

$$N(t) = 200 e^{kt}, \quad k > 0.$$

After 2 hours, the culture contains 800 cells.

- (a) Write down the value of N when $t = 0$.
- (b) Show that $k = \frac{1}{2} \ln 4$.
- (c) Find the number of cells after 5 hours, to the nearest integer.
- (d) Find the time, in hours, at which the number of cells first reaches 5000. Give your answer correct to three significant figures.
- (e) Find the doubling time of the culture. Give your answer correct to three significant figures.

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2. Marco deposits \$3000 in a savings account that pays interest of 4.5% per annum, compounded annually. No further deposits or withdrawals are made.
- (a) Show that the value of the account at the end of the first year is \$3135.
 - (b) Find the value of the account at the end of 6 years, correct to the nearest dollar.
 - (c) Find the smallest number of full years required for the value of the account to first exceed \$5000.
 - (d) Suppose that, instead, the interest is compounded continuously at a nominal annual rate of 4.5%, so that the value after t years is $A(t) = 3000e^{0.045t}$. Find the value of this account at the end of 6 years, correct to the nearest cent, and state the difference between the two accounts at that time.

3. Cesium-137 is a radioactive isotope used in some medical treatments. The mass remaining, in grams, t years after a sample is prepared is modelled by

$$M(t) = M_0 \left(\frac{1}{2} \right)^{t/T},$$

where M_0 is the initial mass and T is the half-life. For cesium-137, $T = 30$ years. A sample is prepared with $M_0 = 64$ grams.

- (a) Find the mass remaining after 50 years. Give your answer correct to three significant figures.
- (b) Find the time at which the mass has decayed to 8 grams.
- (c) Find, to the nearest year, the time after which only 5% of the original mass remains.

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4. A cup of coffee is poured. Its temperature T , in degrees Celsius, t minutes after pouring, is modelled by

$$T(t) = 22 + 64e^{-kt}, \quad k > 0.$$

After 4 minutes, the temperature of the coffee is 54°C .

- (a) State the temperature of the coffee at the moment it is poured.
- (b) State the temperature of the room.
- (c) Show that $k = \frac{1}{4} \ln 2$.
- (d) Find the temperature of the coffee after 10 minutes. Give your answer correct to three significant figures.
- (e) The coffee is comfortable to drink when its temperature reaches 30°C . Find the time at which this occurs.

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